

Intersection of lines

Basic skills

Q1. $\Gamma_1 = \begin{pmatrix} 1+2\lambda \\ -2+3\lambda \end{pmatrix}$ $\Gamma_2 = \begin{pmatrix} \mu \\ 3-\mu \end{pmatrix}$

single vector form

a) $x = 1+2\lambda$, $y = -2+3\lambda$

b) $x = \mu$, $y = 3-\mu$

c) $1+2\lambda = \mu$
 $-2+3\lambda = 3-\mu$
 $\Rightarrow -2+3\lambda = 3-1-2\lambda$
 $\Rightarrow 5\lambda = 4$ $\lambda = \frac{4}{5}$
 $\mu = 1+2 \times \frac{4}{5} = \frac{13}{5}$

d) so using $\mu = \frac{13}{5}$

$x = \frac{13}{5}$, $y = 3 - \frac{13}{5} = \frac{2}{5}$ $\left(\frac{13}{5}, \frac{2}{5}\right)$

e) for Γ_1 points $(1, -2)$, $(3, 1)$

so $y = \frac{3}{2}x - \frac{7}{2}$

Γ_2 points $(0, 3)$, $(1, 2)$

$y = 3 - x$

$$f) \quad \frac{3}{2}x - \frac{7}{2} = 3 - x \quad \frac{5}{2}x = \frac{13}{2} \quad x = \frac{13}{5}$$

$$\text{then } y = 3 - \frac{13}{5} = \frac{2}{5} \quad \text{same as d)}$$

Q2. 3 equations, 2 unknowns

Either $\begin{cases} \text{infinite sols} \\ 1 \text{ unique} \\ \text{no sols} \end{cases}$

Competency skills

$$Q3. a) \quad \underline{\Gamma}_1 = \begin{pmatrix} 2+\lambda \\ 5\lambda \\ 5+2\lambda \end{pmatrix} \quad \underline{\Gamma}_2 = \begin{pmatrix} 7-2\mu \\ 7-\mu \\ 1+3\mu \end{pmatrix}$$

$$\text{for } \underline{\Gamma}_1 = \underline{\Gamma}_2 \quad x: 2+\lambda = 7-2\mu$$

$$y: 5\lambda = 7-\mu$$

$$z: 5+2\lambda = 1+3\mu$$

$$b) \quad \lambda = 5-2\mu \text{ from } x \text{ then}$$

$$y: 25-10\mu = 7-\mu \quad 9\mu = 18 \quad \mu = 2$$

$$\text{then } \lambda = 1$$

c) for $z: 5 + 2 \times 1 = 1 + 3 \times 2$

$$7 = 7 \text{ yes}$$

So intersection when $\lambda = 1, \mu = 2$

$$\text{So } \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 3 \\ 5 \\ 7 \end{pmatrix}$$

d) As z changes to $1 + 4\mu$ for line two no longer consistent as

$$5 + 2 \times 1 \neq 1 + 4 \times 2$$

So no intersection, lines skew as not parallel

Q 4. a) $\underline{r}_1 = \begin{pmatrix} 2 + 2\lambda \\ 3 + 2\lambda \\ 2 - 4\lambda \end{pmatrix} \quad \underline{r}_2 = \begin{pmatrix} -1 + \mu \\ -3 + 2\mu \\ -1 + \mu \end{pmatrix}$

$$\text{So } \begin{aligned} 2 + 2\lambda &= -1 + \mu & \text{so } \mu &= 3 + 2\lambda \\ 3 + 2\lambda &= -3 + 2\mu \end{aligned}$$

$$\text{So } \mu = -3 + 2\mu \Rightarrow \mu = 1 \Rightarrow \lambda = -1$$

then check $z: 2 - 4 \times -1 \neq -1 + 1$ so no point

$$b) \quad \frac{x-6}{3} = \frac{y-3}{2} = \frac{z}{4} = \lambda$$

$$\frac{x+2}{2} = \frac{y}{3} = \frac{z+0.4}{10} = \mu$$

$$x = 3\lambda + 6 = 2\mu - 2$$

$$y = 2\lambda + 3 = 3\mu \quad \text{so} \quad \mu = \frac{2}{3}\lambda + 1$$

$$\Rightarrow 3\lambda + 6 = \frac{4}{3}\lambda$$

$$\text{so} \quad \frac{5}{3}\lambda = -6 \quad \lambda = -\frac{18}{5} \Rightarrow \mu = -\frac{7}{5}$$

$$\text{then} \quad z = 10 \times -\frac{7}{5} - 0.4 = -\frac{72}{5}$$

$$z = 4 \times -\frac{18}{5} = -\frac{72}{5}$$

Consistent so intersect at

$$x = 2 \times -\frac{7}{5} - 2 \quad y = 3 \times -\frac{7}{5}$$

$$= -\frac{24}{5} \quad y = -\frac{21}{5} \quad z = -\frac{72}{5}$$

$$\left(-\frac{24}{5}, -\frac{21}{5}, -\frac{72}{5}\right)$$



Q5. $\Gamma_1 = \begin{pmatrix} 3 \\ -2 \\ 1 \end{pmatrix} + \lambda \begin{pmatrix} 6 \\ 0 \\ 5 \end{pmatrix}$ $\Gamma_2 = \begin{pmatrix} 2 \\ 2 \\ 0 \end{pmatrix} + \mu \begin{pmatrix} 5 \\ 1 \\ 1 \end{pmatrix}$

clearly not parallel as $\begin{pmatrix} 6 \\ 0 \\ 5 \end{pmatrix} \neq k \begin{pmatrix} 5 \\ 1 \\ 1 \end{pmatrix}$ for any k

now

$$3 + 6\lambda = 2 + 5\mu$$

$$-2 = 2 + \mu$$

$$\Rightarrow \mu = -4$$

$$\Rightarrow 3 + 6\lambda = -18$$

$$6\lambda = -21$$

$$\lambda = -\frac{7}{2}$$

so $1 + 5\lambda = 1 + 5 \times -\frac{7}{2} = -\frac{33}{2}$

$\mu = -4$ so inconsistent

so lines are skew

Q6. Lines are

$$\Gamma_1 = \begin{pmatrix} -5 \\ 2 \\ -2 \end{pmatrix} + \lambda \begin{pmatrix} 10 \\ 1 \\ 4 \end{pmatrix} \quad \Gamma_2 = \mu \begin{pmatrix} -3 \\ 2 \\ k \end{pmatrix}$$

diff of coordinates

$$0 \leq \lambda, \mu \leq 1$$

on string

For intersection :

$$-5 + 10\lambda = -3\mu$$

$$2 + \lambda = 2\mu \quad \lambda = 2\mu - 2$$

$$\text{SO } -5 + 20\mu - 20 = -3\mu$$

$$23\mu = 25 \quad \mu = \frac{25}{23}$$

$$\lambda = \frac{4}{23}$$

$$\text{so need } -2 + 4\lambda = k\mu$$

$$\text{i.e. } k = \frac{-2 + 4 \times \frac{4}{23}}{\frac{25}{23}} = -\frac{6}{5}$$

a) NO μ too high

b) NO μ too high

c) NO such k as μ is too high
even though $k = -\frac{6}{5}$ would be
consistent but point of intersection
is off the blue string